

Different algorithms used

1. Linear Regression

Linear Regression is a supervised machine learning technique.

Linear Regression model tries to fit the best fit line between the independent and dependent variables.

Linear regression = min(Sum of squared errors)

2. Elastic Net

Elastic Regression is a regularized machine learning technique, that consists of penalties from both the lasso and ridge regression to obtain the best output.

Elastic Net regression = min(Sum of squared errors)+ (alpha1 *| slope|) + (alpha * slope)square)

3. Ridge Regression (L2 Regularization)

Ridge Regression is a further extension of linear regression where a penalty is added to the RSS (residual sum of squares) which is equal to the square of coefficients term.

This penalty term is nothing but alpha multiplied by slope and its square. This helps getting rid of overfitting.

Ridge regression = min(Sum of squared errors + alpha * slope)square)

4. Lasso Regression (L1 Regularization)

Lasso Regression is similar to ridge regression, the only difference is the penalty term.

Here, the alpha value is multiplied by the absolute value of the slope.

Lasso Regression = min(sum of squared error + alpha * | slope|)

5. XGBoost Regressor

Extreme Gradient Boosting is an ensemble learning technique. It has an enhanced speed and performance in training sequential decision trees i.e the errors made by the previous models, are improved by the succeeding models.

6. AdaBoost Regressor

Adaboost is an ensemble technique that is built of decision trees formed sequentially. Adaptive boosting (Adaboost) weights are assigned to each instance with higher weights assigned to high error instances.

7. Gradient Boosting Regressor

Gradient Boosting is an ensemble learning technique that combines several base learners to form one optimal model. The aim of gradient boosting is to minimize the loss function by adding weak learners using gradient descent.

8. Decision Tree Regressor

Decision trees models are built in the form of the tree structures. The dataset is broken down into smaller subsets and the final result is a tree with decision nodes.

9. Random Forest Regressor

Random Forest is an ensemble machine learning technique that uses multiple decision trees (base learners) to built a final optimal model.